

# Tom de Groot

**Independent Researcher**
**AI for Math & Science**

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## METHODS & TOOLS

**Research:** agentic workflows, evaluations, formal verification

**Code:** Python, Lean 4, PyTorch, SQL, RDKit

**Systems:** Linux/HPC, AWS, Docker, Git

**Languages:** Dutch, English

## EDUCATION

**BSc Chemistry**
University of Groningen
2021 - 2025
*Track: Chemistry of Life*

**Thesis:** GNN evaluation for NMR prediction

## RECOGNITION

Harmonic Rising Mathematician

Chess.com Member of the Year 2021

# INDEPENDENT RESEARCHER

**AI-ASSISTED MATHEMATICS | FORMAL VERIFICATION | SCIENTIFIC ML**

I use frontier models to attack open problems, then make the result inspectable: expert review, formal proof, and reproducible code.

> **"the gold standard of LLM-aided mathematics"**
> Daniel Litt on Tom's Lean-verified FirstProof solution

## SELECTED RESEARCH

**Daniel Litt Open Problem - Raynaud Bundles** — Jul 2026
Steered GPT-5.5 Pro and GPT-5.6 Sol Pro from a partial result to a complete candidate proof using Raynaud bundles. Now auditing the argument with Daniel Litt's feedback. Research note | Proof thread

**FirstProof Problem 9 - Lean-Verified** — Feb 2026
Orchestrated LLM agents from informal search to Lean 4 verification. The repository's only solved problem. Proof + code

**Prime Number Theorem And (PNT+)** — 2025 - Present
Contributed merged Lean proofs; use compilation and adversarial review to test model-generated mathematics. Merged PRs

## SCIENTIFIC AI SYSTEMS

**Bazaar_ML - Simulator, Certification & RL** — 2026 - Present
**49,248-case** executable certification matrix; **99.85%** sampling-pool coverage; deterministic traces, provenance, and agent-training hooks.

**NMR Prediction with Graph Neural Networks** — 2024 - 2025
Curated molecular/NMR data and benchmarked GNNs on HPC under controlled splits and evaluation.

## ENGINEERING

**AI Engineer - SoundImports** — Present
Ship prediction algorithms, AI automation, and containerized AWS services.

**BaanSoliciteren - Agentic Job Automation** — 2025 - 2026
Auditable pipeline for vacancy ingestion, fit scoring, and application-state tracking.
GitHub

**Earlier: Teamspot (Regional Director / QA), Transcom, Chess.com** — 2022 - 2024